TitraHealth

The Complete GLP-1 Medication Companion

Product Feature Overview

Version 1.0 · March 2026

Confidential — For Authorized Recipients Only

Table of Contents

1. Introduction — What Is TitraHealth?
2. Getting Started — Authentication & Onboarding
3. Home Dashboard — Your Daily Command Center
4. Logging — The Nine Entry Types
5. Insights & Log Page — Understanding Your Data
6. Weekly Check-Ins — Structured Self-Assessment
7. Weekly Summary — Your Cycle in Review
8. Provider Report — Clinical Communication Tool
9. Intelligence Layer — The Brain Behind the App
10. AI Features — Powered by OpenAI & Anthropic
11. Wearable Integration — Connected Health
12. Settings & Notifications
13. Design System — Look and Feel

1. Introduction — What Is TitraHealth?

The Problem

GLP-1 receptor agonist medications — brands like **Ozempic**, **Wegovy**, **Mounjaro**, **Zepbound**, **Saxenda**, and **Rybelsus** — have transformed weight management. But the medication alone is only part of the equation. Clinical research shows that **26–40% of weight lost on GLP-1s is lean muscle mass**, that **60% of lost weight is regained within 18 months of stopping**, and that behavioral habits formed *during* treatment are the strongest predictor of long-term success.

Generic fitness and diet apps were not designed for this population. They don't understand injection cycles, medication phases, titration schedules, GLP-1–specific side effects, or the pharmacokinetic curves that determine when appetite suppression peaks and fades. Patients are left to manage a complex medical therapy with tools that weren't built for them.

The Solution

TitraHealth is a purpose-built companion app for people on GLP-1 medications. It ties together medication adherence, nutrition tracking, activity monitoring, side-effect logging, wearable biometrics, weekly self-assessments, and AI-powered coaching into a single, unified experience. Every feature is designed with the specific needs and challenges of GLP-1 patients in mind.

> **WHY THIS MATTERS FOR GLP-1 PATIENTS**
>
> GLP-1 medications create a limited "plasticity window" — a period of reduced appetite and metabolic change — during which patients can build sustainable habits. TitraHealth helps patients maximize this window by providing real-time, phase-aware guidance that adapts to their medication cycle, side effects, and biometric responses.

Supported Medications

DRUG CLASS	BRAND NAMES	ROUTE	FREQUENCY
Semaglutide	Ozempic, Wegovy	Subcutaneous injection	Weekly
Tirzepatide	Mounjaro, Zepbound	Subcutaneous injection	Weekly
Liraglutide	Saxenda, Victoza	Subcutaneous injection	Daily
Dulaglutide	Trulicity	Subcutaneous injection	Weekly
Oral Semaglutide	Rybelsus	Oral tablet	Daily
Orforglipron	Orforglipron (investigational)	Oral tablet	Daily

Platform

TitraHealth is built with **React Native** and **Expo**, supporting both **iOS** and **Android**. The backend is powered by **Supabase** (PostgreSQL database with row-level security, authentication, and real-time capabilities). AI features use **OpenAI** (GPT-4o) and **Anthropic** (Claude) models.

2. Getting Started — Authentication & Onboarding

2.1 Sign-In Options

Users can access TitraHealth in three ways:

- **Email & Password** — Traditional registration and login with password validation (6+ characters). Includes “Remember me” and password recovery options.
- **Google Sign-In** — One-tap OAuth authentication using Google accounts, implemented via secure PKCE flow.
- **Demo Mode** — Allows full app exploration without creating an account, loading a sample profile with mock data so prospective users can experience every feature before committing.

2.2 Guided Onboarding

After signing up, new users complete a guided onboarding flow that collects the information needed to personalize every aspect of the app. This includes:

- **Medication details** — Drug type, brand, current dose, and injection frequency
- **Body measurements** — Height, current weight, sex, and birth year
- **Goals** — Target weight and desired weekly weight-loss rate
- **Activity level** — Sedentary, lightly active, moderately active, or very active
- **Program start date** — When the user began their GLP-1 medication

> **WHY THIS MATTERS**
>
> Every downstream feature — daily nutrition targets, PK-curve modeling, injection phase detection, clinical alerts, AI context, and weight projections — depends on accurate onboarding data. The onboarding flow is designed to be quick (under 2 minutes) while capturing everything needed for deep personalization.

3. Home Dashboard — Your Daily Command Center

The home dashboard is the first screen users see when they open the app. It provides a comprehensive, at-a-glance view of their current medication status, daily progress, and any items requiring attention.

3.1 Interactive Calendar

A month-view calendar strip at the top of the dashboard displays the current month with color-coded day indicators:

- **Orange dot** Injection day — a dose was logged on this day
- **Green dot** Logging day — food, activity, or other entries were logged
- **Blue dot** Today's date

Tapping any past date loads a historical snapshot of that day's complete log summary, allowing users to review what they ate, how they felt, and what they logged on any previous day. Dates before the program start date are disabled.

3.2 Medication Banner

A prominent banner shows the user's current medication brand name (e.g., "Mounjaro"), their current dose (e.g., "2.4 mg"), and which program week they are in. This provides instant orientation for both the user and anyone glancing at their screen during a provider visit.

3.3 Treatment Progress Card

For injectable medications, this card shows the current **injection phase**:

- **Shot Day** — The day the injection was administered
- **Peak Effect** — When the medication reaches maximum blood concentration
- **Active / Balance** — The period of sustained therapeutic effect
- **Fading / Reset** — When the medication is wearing off before the next dose

For daily oral medications (Rybelsus, orforglipron), the card shows intraday phases: Recently Dosed, Peak Effect, and Approaching Trough. Each phase includes specific guidance text tailored to what the user should expect and focus on during that phase.

3.4 Daily Log Summary

An expandable card that displays everything logged today. In its compact view, it shows summary icons and counts. When expanded, it reveals full detail with **inline edit modals** for:

- **Food entries** — Edit name, calories, macros (protein/carbs/fat/fiber), and meal type
- **Activity entries** — Edit exercise type, duration, steps, and active calories

- **Weight** — Edit the logged weight value
- **Injection details** — View medication, dose, and injection site
- **Side effects** — View logged symptoms with severity badges

A **Metrics Impact** section shows progress bars for daily targets: calories, protein, fiber, and water, each displayed as current intake versus the personalized daily target.

3.5 Appetite Forecast Strip

A visual strip that shows the predicted appetite suppression level for the coming days, driven by **pharmacokinetic modeling**. Based on when the last injection was administered and the known drug concentration curve, the app forecasts when appetite suppression will be strongest and when it will fade. For daily medications, this is broken into 6 four-hour intraday blocks.

> **WHY THIS MATTERS**
>
> Understanding when appetite suppression peaks and fades helps patients plan meals strategically — eating protein-dense meals when appetite is lowest (during peak effect) and preparing for increased hunger during the fading phase. This is unique to GLP-1 medication management and is not available in any generic health app.

3.6 Clinical Alert Cards

Evidence-based alerts appear automatically based on the user's program week, logged data, and clinical guidelines:

- **Lab reminders** — Iron/ferritin at week 8, Vitamin D at week 12
- **Hair loss reassurance** — Educational messaging during weeks 12–26 (a common but temporary side effect)
- **Plateau detection** — Coaching protocol if weight has been stable for more than 2 weeks
- **Lean mass alert** — Warning if weight loss is too rapid (risk of muscle loss)
- **Resistance training nudge** — Recommendation to add strength training for lean mass preservation
- **Dropout risk** — Prompt if there are long gaps in logging activity

3.7 Weekly Check-In Prompt

Seven days after the last check-in, a prominent card appears encouraging the user to complete their weekly wellness surveys. These structured self-assessments track trends in GI symptoms, energy, mood, appetite, food noise, sleep quality, and mental health.

3.8 Injection Countdown

A real-time countdown displays how many days remain until the next scheduled injection, with overdue detection and visual emphasis when a dose is missed. For daily medications, it shows time since the last

dose.

4. Logging — The Nine Entry Types

4.0 The Floating Action Button & Entry Sheet

A persistent orange **Floating Action Button (FAB)** sits at the bottom-right of every main screen. Tapping it opens a frosted-glass bottom sheet displaying a 3×3 grid of all entry types, each with a distinct icon and label. The sheet also includes an **inline water logging** feature with quick-tap amount buttons and unit conversion (mL or oz based on user preference).

The nine entry types in the grid are:

Describe Food | Log Injection | Capture Food | Scan Food | Ask AI | Side Effects | Log Weight | Log Water | Log Activity

---

4.1 Log Weight

A custom-built **interactive ruler picker** lets users set their weight by dragging horizontally. The ruler displays 0.1-unit tick marks with integer labels, and provides **haptic feedback** every 0.5 units for a tactile feel. The current weight is shown in a large orange display (72pt font).

- **Unit toggle** — Switch between pounds (lbs) and kilograms (kg) with automatic conversion
- **Voice input** — Say “185 pounds” or “84 kilograms” and the value auto-fills
- **HealthKit sync** — On save, the weight is written to Apple Health (iOS) or Health Connect (Android)

4.2 Log Injection

Captures every aspect of a medication injection:

- **Medication selector** — Pre-filled from the user’s profile; shows brand name (Ozempic, Mounjaro, etc.)
- **Dose selector** — Horizontal scroll of brand-specific dose chips (e.g., 0.25, 0.5, 1.0, 1.7, 2.4 mg for semaglutide)
- **Injection site grid** — 6-site visual grid (Left/Right Abdomen, Left/Right Thigh, Left/Right Upper Arm) with a “Rotate To” suggestion based on the last 5 injection sites
- **Fasting window** — For oral drugs, a toggle to confirm the 30-minute empty-stomach requirement
- **Batch number** — Optional field to record the medication batch (for quality tracking)
- **Notes** — Free-text field with voice input support

> **WHY THIS MATTERS**
>
> Injection site rotation is medically important — repeated injection at the same site can cause lipodystrophy (abnormal fat deposits) that impair drug absorption. TitraHealth automatically tracks the last 5 sites and suggests the optimal next site.

4.3 Log Food — The Unified Food Hub

The food logging screen is a comprehensive hub with **four input modes**, accessible via mode pills in the header:

Search Mode

A debounced search bar queries the **FatSecret food database** (2.3 million foods including restaurant chains and real serving sizes) via a secure server-side proxy. Results display food name, brand, calories, and macros. When the database returns no results, an **AI fallback** estimates nutrition using GPT-4o.

Scan Mode

An inline barcode scanner (using the device camera) reads UPC/EAN barcodes and looks up the product in the **Open Food Facts** database. Recognized products display their full nutrition information with configurable serving sizes.

Describe Mode

Users type a natural-language description of what they ate (e.g., “2 scrambled eggs with avocado toast and a glass of orange juice”). The AI parser breaks this into individual food items, estimates portion sizes in grams, and looks up each item’s nutrition. Users can select which parsed items to log via checkboxes.

Camera Mode

Users photograph their food, and **GPT-4o Vision** analyzes the image to identify food items and estimate their nutritional content.

Shared Features Across All Modes

- **Meal Tray** — A multi-item assembly area where foods are queued before logging. Users can add items from any mode, adjust serving sizes, and see running totals for calories and protein.
- **Serving Size Selection** — Scrollable serving options (e.g., “1 cup”, “1 medium”, “100g”) or custom gram input. Nutrition is automatically recalculated.
- **Meal Type Selector** — Tag entries as Breakfast, Lunch, Dinner, or Snack
- **Saved Meals** — Save frequently eaten combinations as templates for one-tap re-logging
- **Recent Foods** — Quick access to the last 10 foods logged, with favorite marking
- **Custom Foods** — Create a personal food database with custom nutrition values for items not in any database

> **WHY THIS MATTERS**
>
> GLP-1 patients need to prioritize protein intake (to preserve lean mass) and monitor fiber and hydration (common side effects include GI disturbances). Having four flexible input modes ensures that no matter how the user eats — a restaurant meal, home cooking, a packaged snack, or a complex plate — they can accurately log their nutrition in seconds.

4.4 Log Activity

Captures exercise and physical activity with intelligent calorie estimation:

- **Exercise type picker** — 9 preset types: Walking, Running, Cycling, Strength, HIIT, Yoga, Pilates, Swimming, Other. Each has an assigned **MET (Metabolic Equivalent of Task)** value.
- **Duration slider** — 0–120 minutes with haptic feedback, large numeric display
- **Intensity slider** — 1–10 scale with derived labels (Easy, Moderate, Hard)
- **Auto-calculated fields**:
  - Estimated steps = exercise type's steps-per-minute × duration
  - Active calories = MET value × user weight × (duration / 60) × intensity multiplier

4.5 Side Effects

A customizable side-effect tracker designed specifically for the GLP-1 side-effect profile:

- **Predefined effects** — Nausea, vomiting, constipation, diarrhea, acid reflux, bloating, stomach pain, fatigue, headache, dizziness, appetite suppression, hair loss, palpitations, anxiety
- **Custom effects** — Users can add any additional symptom they want to track
- **Severity sliders** — Each effect has a 0–10 severity scale with smooth drag interaction and haptic feedback
- **Phase selector** — Tag symptoms by injection phase (Shot, Peak, Balance, Reset) to identify phase-related patterns

After logging, an **Impact Viewer** screen shows how the reported side effects dynamically adjust daily targets. For example, if nausea is reported, protein targets may increase (to compensate for reduced food intake), water targets may increase, and activity targets may decrease.

> **WHY THIS MATTERS**
>
> GI side effects are the #1 reason patients discontinue GLP-1 medications. By tracking side effects by severity and injection phase, patients and their providers can identify patterns (e.g., nausea always peaks on days 2–3 post-injection) and make informed decisions about dose titration timing.

4.6 Ask AI

A context-aware AI chat interface that acts as a personal health companion. Before every conversation, the app builds a **context snapshot** that includes:

- User demographics and medication details
- Current injection phase and PK concentration
- Today's nutrition versus targets
- Recent side effects (7-day window)
- Weekly check-in scores
- Weight progress and BMI
- Wearable biometrics (HRV, heart rate, sleep)

This context is injected into the AI's system prompt, enabling highly personalized responses. Suggestion chips on the empty state help users start conversations: "Why am I more hungry today?", "How's my protein intake?", "What should I eat tonight?"

4.7 Water Logging

Built directly into the Add Entry Sheet (no separate screen needed). Users tap quick-add amount buttons, with automatic conversion between milliliters and ounces based on their unit preference. Includes AI-powered text parsing for natural-language entries (e.g., "2 glasses of water").

4.8 Voice Input

Available on **five logging screens** (weight, injection, activity, side effects, and food describe mode). The flow is:

1. User taps the microphone button (pulsing animation indicates recording)
2. Audio is captured and sent to **OpenAI Whisper** for transcription
3. The transcribed text is parsed by a **context-aware AI model** that understands which screen the user is on and extracts structured data accordingly
4. Parsed values auto-fill the form fields

For example, on the injection screen, saying "Ozempic 1 milligram, left thigh, batch number ABC123" auto-fills the medication, dose, site, and batch fields.

5. Insights & Log Page — Understanding Your Data

The Insights page is divided into three tabs — **Medication**, **Lifestyle**, and **Progress** — each providing a different lens on the user's health journey.

5.1 Medication Tab

Pharmacokinetic (PK) Curve Chart

A smooth SVG chart that visualizes the estimated drug concentration in the body over time, modeled using the **Bateman equation** (the same pharmacokinetic model used in FDA drug-approval studies). The chart shows a 28-point smooth curve with gradient fill, a real-time "NOW" marker showing current estimated concentration, and labeled peak and trough levels.

Timeline Log Entries

A chronological feed of all logged entries for the current period, showing food items (with calories, protein, fiber), activity (type, duration, steps), injections (medication, dose, site, next site suggestion), and weight entries (with delta from previous log).

Health Monitor Cards

Six to eight biometric cards display real-time data from connected wearable devices:

METRIC	SOURCE	WHAT IT SHOWS
Resting Heart Rate	Apple Health / Garmin	BPM with gauge bar, status badge (normal / elevated / low)
Heart Rate Variability	Apple Health / Garmin	ms SDNN with status interpretation
Sleep Duration	Apple Health / Garmin	Hours and minutes, quality assessment
SpO2 (Oxygen Saturation)	Apple Health	Percentage with range indicator
Temperature	Apple Health	Deviation from baseline
Blood Glucose	Apple Health (CGM data)	mg/dL with range classification (if available from Dexcom, Libre, or Stelo via Apple Health)

Each card features a **gauge bar** visualization showing where the value falls within its expected range. **Long-pressing** any card opens an AI chat pre-loaded with that metric's context, asking questions like "Why is my HRV lower than usual?"

> **WHY THIS MATTERS**
>
> Research shows that GLP-1 medications measurably decrease HRV (–6.2ms SDNN at 12 weeks) and increase resting heart rate. By tracking these biometrics in the context of the medication cycle, TitraHealth can distinguish expected pharmacological effects from concerning deviations that warrant medical attention.

5.2 Lifestyle Tab

- **Metric selector chips** — Toggle between Protein, Carbs, Fat, Fiber, Calories, Steps, and Active Calories
- **Line charts** — Smooth bezier-curve charts with gradient fill, goal-line overlay, and labeled axes
- **Period selector** — View trends over 7 days, 14 days, 30 days, 90 days, or all time
- **Trend analysis** — Compares current period to previous period with percentage change indicators

5.3 Progress Tab

- **Weight loss curve** — Interactive chart showing actual weight over time versus projected trajectory
- **Goal progress ring** — Circular progress indicator showing percentage toward goal weight with days elapsed
- **BMI tracking** — Current BMI calculation with classification (Underweight, Normal, Overweight, Obesity I–III)

6. Weekly Check-Ins — Structured Self-Assessment

TitraHealth includes a comprehensive weekly check-in system covering **seven wellness domains**, each designed to capture the specific challenges and experiences of GLP-1 patients.

6.1 The Seven Domains

DOMAIN	QUESTIONS	WHAT IT MEASURES
GI Burden	3	Nausea severity, abdominal pain, disruption to daily activities from GI symptoms
Energy & Mood	3	Energy levels, focus/concentration, emotional stability
Appetite	3	Difficulty eating enough, food appeal, sense of appetite control
Food Noise	3	Intrusive food thoughts, craving intensity, ability to focus on non-food tasks
Sleep Quality	3	Falling asleep, staying asleep, feeling refreshed, daytime impact
Activity Quality	3	Exercise capacity, strength/endurance, progress toward fitness goals
Mental Health	5	Depression (PHQ-based), anxiety (GAD-based), impact on eating/adherence

6.2 Scoring System

Each question uses a **DotScale** input (0–4 circles) with labels ranging from “Not at all” to “Always/Extremely.” Individual question scores are summed per domain and normalized to a 0–100 scale. Scores are inverted where appropriate (for domains where higher raw scores indicate worse outcomes) so that a higher normalized score always means better wellness.

Each domain produces a **status interpretation** with color coding:

- **70–100** Minimal / Good
- **50–69** Mild / Moderate
- **30–49** Moderate / Notable
- **0–29** Significant / Severe

6.3 Mental Health Screening

The mental health survey uses clinically validated question structures based on the **PHQ-2** (depression) and **GAD-2** (anxiety) screening tools. When a user's score exceeds the clinical threshold (15+ out of 20), a **provider prompt banner** appears encouraging them to discuss their symptoms with a healthcare professional.

> **WHY THIS MATTERS**
>
> "Food noise" — the constant, intrusive thoughts about food — is a hallmark symptom that GLP-1 medications specifically reduce. Tracking it weekly provides a unique, quantifiable measure of medication effectiveness that goes beyond the scale. No other consumer health app tracks food noise as a discrete metric.

6.4 Check-In Summary

After completing a check-in, a summary screen displays the calculated score, interpretation, and domain status. It offers AI chat access with the check-in context pre-loaded, and schedules a reminder for the next weekly check-in.

7. Weekly Summary — Your Cycle in Review

At the end of each 7-day period, TitraHealth generates a comprehensive weekly summary that reviews the user's complete cycle.

7.1 AI-Generated Narrative

An AI-generated 3–4 sentence recap of the week, personalized to the user's actual data. For example: "You averaged 92g of protein daily this week, hitting your target 5 out of 7 days. Your nausea symptoms decreased compared to last week, suggesting your body is adjusting to the current dose."

7.2 Key Metrics

- **Weight change** — Start versus end weight with delta and trending indicator
- **Nutrition averages** — Progress bars for Calories, Protein, Fiber, and Water showing averages versus personalized targets
- **Activity grid** — A 7-day (M–S) visual grid showing active versus inactive days, plus average daily steps
- **Check-in score pills** — Color-coded pills for each of the 7 wellness domains
- **Side effect summary** — Total logged count grouped by symptom type

7.3 PK Model Comparison

When injection logs are present, the summary compares the **predicted appetite suppression** (from the pharmacokinetic model) against the **reported appetite score** (from check-ins). A delta label indicates whether the user's experience "matched expectations," was "higher than predicted," or "lower than predicted." For daily medications, an adherence streak is also displayed.

7.4 PDF Export

The entire weekly summary can be exported as a clean, formatted PDF document for personal records or sharing with a healthcare provider. The PDF includes the AI insight, nutrition table, activity summary, and check-in scores.

8. Provider Report — Clinical Communication Tool

TitraHealth generates professional PDF reports designed for healthcare providers. These reports aggregate clinical data into a format that physicians and nurse practitioners can quickly review during an appointment.

8.1 Configuration

- **Date range presets** — Last 30 days, Last 90 days, or Since Program Start
- **Toggleable sections** — Users can include or exclude specific sections to tailor the report
- **Provider information** — Optional fields for provider name and practice/clinic name
- **AI summary toggle** — Option to include an AI-generated clinical narrative

8.2 Report Contents

SECTION	WHAT IT CONTAINS
Patient Summary	Demographics, BMI (with classification), program week, current weight vs. goal
Medication Summary	Drug type, brand, current dose, route, frequency, initial dose
Weight Trend	SVG line chart with data points, goal weight dashed line, weekly rate, flags for rapid loss or underweight
Medication Adherence	Expected vs. logged doses, adherence percentage, missed dose windows, dose calendar (green = logged, red = missed)
Side Effect Profile	Breakdown by type with count, average/max severity, trend direction. Severity distribution (mild/moderate/severe). GI symptom cluster detection. Weekly trend bars.
Nutrition Summary	Logging compliance, averages vs. targets for calories/protein/carbs/fat/fiber, weekly trend table
Activity & Exercise	Average daily steps vs. target, active days, exercise sessions grouped by type, weekly step trend
Biometrics	Resting HR, HRV, sleep hours, SpO2 from connected wearables
Check-In Trends	Latest scores for all 7 wellness domains, food noise history over time
Clinical Flags	Automated warnings: weight flags, adherence flags, side effect clusters, nutrition concerns, lab reminders, dose titration suggestions

WHY THIS MATTERS

GLP-1 medication management requires regular provider check-ins to evaluate side effects, adjust doses (titration), and monitor for complications. The provider report eliminates the need for patients to recall symptoms from memory during appointments, and gives clinicians structured, longitudinal data to inform clinical decisions. The clinical flags automatically highlight concerns that might otherwise be missed.

8.3 PDF Generation & Sharing

Reports are generated as print-friendly HTML with inline SVG charts, then converted to PDF on-device. Users can share the PDF directly via email, AirDrop, messaging apps, or any share-sheet destination.

9. Intelligence Layer — The Brain Behind the App

Behind TitraHealth's simple interface is a sophisticated intelligence system that continuously processes user data to deliver personalized, clinically informed guidance.

9.1 Dynamic Daily Targets

Daily nutrition and activity targets are not static — they are recalculated continuously based on multiple inputs:

- **Base calculation** — Mifflin-St Jeor BMR equation using height, weight, age, sex, and activity level
- **Medication adjustments** — Targets shift based on whether the user takes injectable or oral GLP-1s
- **Side-effect adjustments** — Active side effects modify targets in real time (e.g., nausea increases protein target to compensate for reduced intake, GI symptoms increase water target)
- **Check-in adjustments** — Weekly survey scores layer additional modifications:
  - High food noise → increase protein and fiber targets
  - Over-suppressed appetite → calorie floor guard (minimum 1,200 kcal)
  - Low energy → reduce activity targets
  - Poor sleep → boost carbohydrate targets
- **Conflict resolution** — When multiple adjustments conflict, the system applies MAX for protein, MAX for fiber, MIN for calories (safety floor), and SUM for steps

> **WHY THIS MATTERS**
>
> A patient experiencing severe nausea on day 2 post-injection has fundamentally different nutritional needs than the same patient on day 6 when symptoms have resolved. Static targets from generic apps fail to account for these medication-driven fluctuations. TitraHealth's dynamic system adapts in real time, always providing achievable, medically appropriate targets.

9.2 Cycle Intelligence

The cycle intelligence engine interprets all user data within the context of their medication cycle:

- **Injection phase detection** — Automatically determines whether the user is in the Shot, Peak, Balance, or Reset phase based on days since injection and drug half-life
- **PK concentration modeling** — Calculates estimated drug concentration at any point in time using the Bateman equation with drug-specific absorption and elimination constants
- **Appetite suppression forecasting** — Predicts daily appetite levels for the coming week based on the PK curve
- **Biometric deviation detection** — Uses Exponential Moving Average (EMA) baselines to distinguish expected medication effects from concerning deviations. For example, a 5ms HRV decrease during peak

phase is expected; a 15ms decrease during balance phase may warrant attention.

- **Intraday blocks** — For daily medications, divides the day into 6 four-hour windows with different expected drug effects

9.3 Escalation Phase Tracking

TitraHealth maps the user's current dose and program week to a clinical **escalation phase**:

PHASE	TYPICAL WEEKS	BEHAVIORAL EMPHASIS
Initiation	1–4	Habit building, baseline tracking
Low Therapeutic	5–8	Protein focus, activity introduction
Mid Therapeutic	9–12	Peak plasticity window — maximum habit formation opportunity
High Therapeutic	13–16	Optimization, sustainability planning
High+	17–20	Maintenance transition
Max Dose	21+	Long-term sustainability

Each phase provides specific behavioral guidance and weekly focus recommendations. The system detects **accelerated** and **slow titrators** and adjusts accordingly.

9.4 Clinical Alerts

Automated, evidence-based alerts monitor for situations that warrant awareness or medical attention:

- Iron/ferritin lab reminder at week 8
- Vitamin D lab reminder at week 12
- Hair loss reassurance messaging (weeks 12–26)
- Weight plateau protocol (stable >2 weeks)
- Rapid weight loss warning (>3 lbs/week)
- Lean mass preservation alert
- Resistance training recommendation
- Dropout risk detection (prolonged logging gaps)
- Thiamine deficiency risk flag (for persistent vomiting)

9.5 Weight Projection

A sigmoid-based projection model estimates when the user will reach their goal weight, factoring in:

- Current loss rate (lbs/week)
- Weight remaining to goal
- Clinical trial benchmarks for the specific drug and dose
- A confidence level: **High** **On Track** **Monitoring**

10. AI Features — Powered by OpenAI & Anthropic

TitraHealth integrates multiple AI models across the entire user experience. AI is not a bolt-on feature; it is woven into logging, insights, coaching, and reporting.

10.1 Food Description Parsing

When users describe a meal in natural language (e.g., “a bowl of chicken tikka masala with naan and a mango lassi”), GPT-4o parses the text into individual food items with estimated serving sizes, calorie counts, and full macro breakdowns (protein, carbs, fat, fiber). Each parsed result includes a **confidence score** (high, medium, low) so users know how reliable the estimate is.

10.2 Voice Log Parsing

After Whisper transcribes spoken audio to text, a context-aware AI model parses the transcription differently depending on which screen the user is on:

- **Weight screen** → Extracts numeric value and unit (lbs/kg)
- **Injection screen** → Extracts medication name, dose, site, batch number, notes
- **Activity screen** → Extracts exercise type, duration, intensity
- **Side effects screen** → Extracts symptom types and severity levels

10.3 Photo Food Analysis

Users can photograph their plate, and **GPT-4o Vision** analyzes the image to identify food items, estimate portion sizes, and calculate approximate nutrition values.

10.4 Dynamic Health Insights

Three categories of AI-generated insights are refreshed daily:

- **Lifestyle insights** — Protein and hydration gaps, fiber intake patterns
- **Medication insights** — Injection cycle status, adherence observations
- **Progress insights** — Weight loss trajectory, milestone recognition

Insights are cached per day to avoid redundant API calls and ensure fast loading.

10.5 Weekly AI Summaries

A personalized 3–4 sentence narrative recap of the past 7 days, generated from the full health context (nutrition, activity, side effects, check-ins, biometrics, and medication adherence).

10.6 Health Monitor AI Chat

Long-pressing any health monitor card (HRV, heart rate, sleep, etc.) opens an AI conversation pre-loaded with that metric's recent data and the user's medication context. The AI explains whether observed values are expected for the current injection phase or warrant attention.

10.7 Context Snapshot Builder

Every AI conversation is powered by a dynamically assembled **context snapshot** that includes: user demographics, medication history, injection phase, program duration, weight progress, today's nutrition vs. targets, recent side effects, weekly check-in scores, BMI, and biometric data. This ensures every AI response is deeply personalized rather than generic.

> **AI MODELS USED**
>
> **OpenAI GPT-4o-mini** — Primary model for insights, food parsing, voice parsing, and chat
> **OpenAI GPT-4o** — Vision model for food photo analysis
> **OpenAI Whisper** — Audio transcription for voice input
> **Anthropic Claude Haiku** — Fast reasoning for specific coaching tasks

11. Wearable Integration — Connected Health

TitraHealth connects to the user's health devices to enrich its intelligence with objective biometric data, creating a more complete picture than self-reported data alone.

11.1 Apple HealthKit (iOS)

On iOS devices, TitraHealth reads and writes data to Apple Health:

Read from Apple Health

Daily steps, Active calories, Heart Rate Variability (HRV), Resting Heart Rate, Sleep duration, SpO2 (oxygen saturation), Blood glucose (from CGM devices like Dexcom, Libre, Stelo), Body weight, Nutrition data

Write to Apple Health

Weight entries (from log-weight), Nutrition macros (from food logging), Active calories (from activity logging)

11.2 Android Health Connect

On Android devices, TitraHealth integrates with **Health Connect** (Google's unified health data platform), providing the same read/write scope as Apple HealthKit for full feature parity across platforms.

11.3 Garmin Connect

Users with Garmin wearable devices can connect their Garmin account via **OAuth 2.0** (PKCE flow). Data is synced through a cloud-based integration using Supabase Edge Functions:

- Daily steps and active calories
- Sleep duration and quality
- Resting heart rate
- Body weight (from Garmin Scale)

The Garmin integration includes full token management (exchange, refresh, and revocation) and displays the last sync timestamp in settings.

WHY THIS MATTERS

Wearable biometrics are not just nice-to-have data points for GLP-1 patients — they serve as **pharmacodynamic markers**. Changes in HRV, resting heart rate, and sleep patterns correlate with drug concentration and effectiveness. By continuously monitoring these metrics, TitraHealth's Cycle Intelligence engine can detect whether the medication is having its expected physiological effect and alert users (and their providers) when deviations occur.

12. Settings & Notifications

12.1 Profile Management

All onboarding data can be updated at any time through dedicated settings screens:

- **Medication** — Change brand, dose, or medication type (triggers recalculation of all PK models and targets)
- **Body** — Update height and weight (in lbs/kg or ft-in/cm)
- **Goals** — Adjust goal weight and target weekly loss rate
- **Personal** — Update sex, birth year, and activity level

12.2 Appearance

A **light/dark theme toggle** switches the entire app between dark mode (default: black background with orange accents) and light mode (white background with the same orange accent system). The theme preference is persisted across app restarts.

12.3 Notification Reminders

TitraHealth provides a configurable push notification system with a master enable/disable toggle and four reminder categories:

REMINDER TYPE	DEFAULT TIME	PURPOSE
Meals	8:00 AM & 7:00 PM	Remind to log food intake
Weight	7:30 AM	Encourage consistent morning weigh-ins
Side Effects	9:00 PM	Evening reflection on symptoms experienced
Daily Plan	8:00 AM	Morning motivation with daily targets

12.4 Smart Dose Reminders

The notification system is **drug-frequency-aware**. For weekly injectables, dose reminders are scheduled on the expected injection day. For daily oral medications, reminders occur every day. Check-in reminders trigger 7 days after the last completed check-in.

All notifications support **deep linking**, meaning tapping a notification takes the user directly to the relevant logging screen (e.g., a meal reminder opens the food logging hub).

12.5 Device Integrations

Toggle switches for Apple Health and Garmin Connect, with connection status and last sync timestamps.

12.6 Provider Report Access

Direct access to the Provider Report generator from the settings screen.

13. Design System — Look and Feel

TitraHealth uses a cohesive design language built around clarity, accessibility, and a premium aesthetic.

13.1 Visual Identity

ELEMENT	DARK THEME (DEFAULT)	LIGHT THEME
Background	#000000 (pure black)	#FFFFFF (white)
Card Surface	#111111 (dark gray)	#F2F2F7 (light gray)
Primary Text	#FFFFFF	#000000
Secondary Text	#9A9490 (muted)	#6B6868
Accent Color	#FF742A (orange) — consistent across both themes	
Success	#27AE60 (green)	
Warning	#F39C12 (amber)	
Error	#E74C3C (red)	

13.2 Glassmorphism

Throughout the app, cards and overlays use a **glassmorphism** effect — a frosted-glass aesthetic achieved with background blur (intensity 80), semi-transparent overlays, and subtle gradient borders. This creates visual depth and hierarchy while maintaining readability.

13.3 Interaction Design

- **Haptic feedback** — Tactile responses on sliders (every unit change), buttons (light tap), and successful actions (success vibration)
- **Animated tab bar** — The bottom navigation collapses into a mini circle when scrolling and expands back on tap, keeping the interface clean during content consumption
- **Score rings** — Animated circular progress indicators for wellness scores and goal progress
- **Smooth transitions** — Modal screens slide up with spring physics, charts animate on data load

13.4 Typography

The app uses **Helvetica Neue** as its primary typeface with weights ranging from 400 (regular) to 800 (extra bold). Headings use uppercase styling with wide letter-spacing (3.5px) for a clean, clinical aesthetic. Large numeric displays (weight values, scores) use 52–72pt font sizes for instant readability.

13.5 Accessibility

All interactive elements maintain appropriate contrast ratios. The orange accent (#FF742A) on dark backgrounds exceeds WCAG AA standards. Font sizes and touch targets are designed for comfortable use across all screen sizes.

Summary

TitraHealth is a comprehensive, clinically-informed companion app purpose-built for the GLP-1 medication population. It combines:

- **9 logging modalities** (weight, injection, food via 4 modes, activity, side effects, AI chat, water, voice)
- **7 weekly wellness domains** (including the only consumer food-noise tracker)
- **Real-time pharmacokinetic modeling** (FDA-grade Bateman equation drug curves)
- **Dynamic daily targets** (adjusted by side effects, check-ins, and medication phase)
- **Wearable biometric integration** (Apple Health, Health Connect, Garmin)
- **AI-powered coaching** (OpenAI GPT-4o + Anthropic Claude)
- **Provider communication tools** (configurable clinical PDF reports)
- **Evidence-based clinical alerts** (lab reminders, plateau detection, safety flags)

Every feature is designed with one goal: to help GLP-1 patients maximize their treatment window, build sustainable habits, and achieve lasting health outcomes — not just weight loss.

TitraHealth — Confidential — March 2026
For questions or further information, contact the TitraHealth team.